Mid progress report

Multimodal Explainable AI for Early-Stage Skin Cancer Classification

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Repo link:

https://github.com/Ahmed-Mohsen-2005/XAI-skin-cancer-project

1. Project Overview

Models implemented:

- Ahmed Mohsen: ConvNeXt Multimodal, DaViT Multimodal, MaxViT Multimodal
- Abdelrahman Mohamed: DenseNet121, EfficientNet-B3 Cross-Attention, ConvNeXt+FT-Transformer, EfficientNetV2-S Gated
- Jana Ahmed: EfficientNet-B3 Late Fusion
- Malak Mohamed: ViT Early Fusion, Hybrid ViT-MLP

This report documents the mid-progress state of our multimodal deep learning pipeline for binary skin lesion classification (melanoma vs. non-melanoma) on the ISIC-DICM-17K dataset (17,060 dermoscopic images + clinical metadata). Each team member independently implemented, trained, and evaluated one or more deep learning architectures that fuse visual features from pretrained image backbones with structured clinical metadata (age, sex, anatomical site). All models include Explainable AI (XAI) components.

1.1 Dataset & Preprocessing

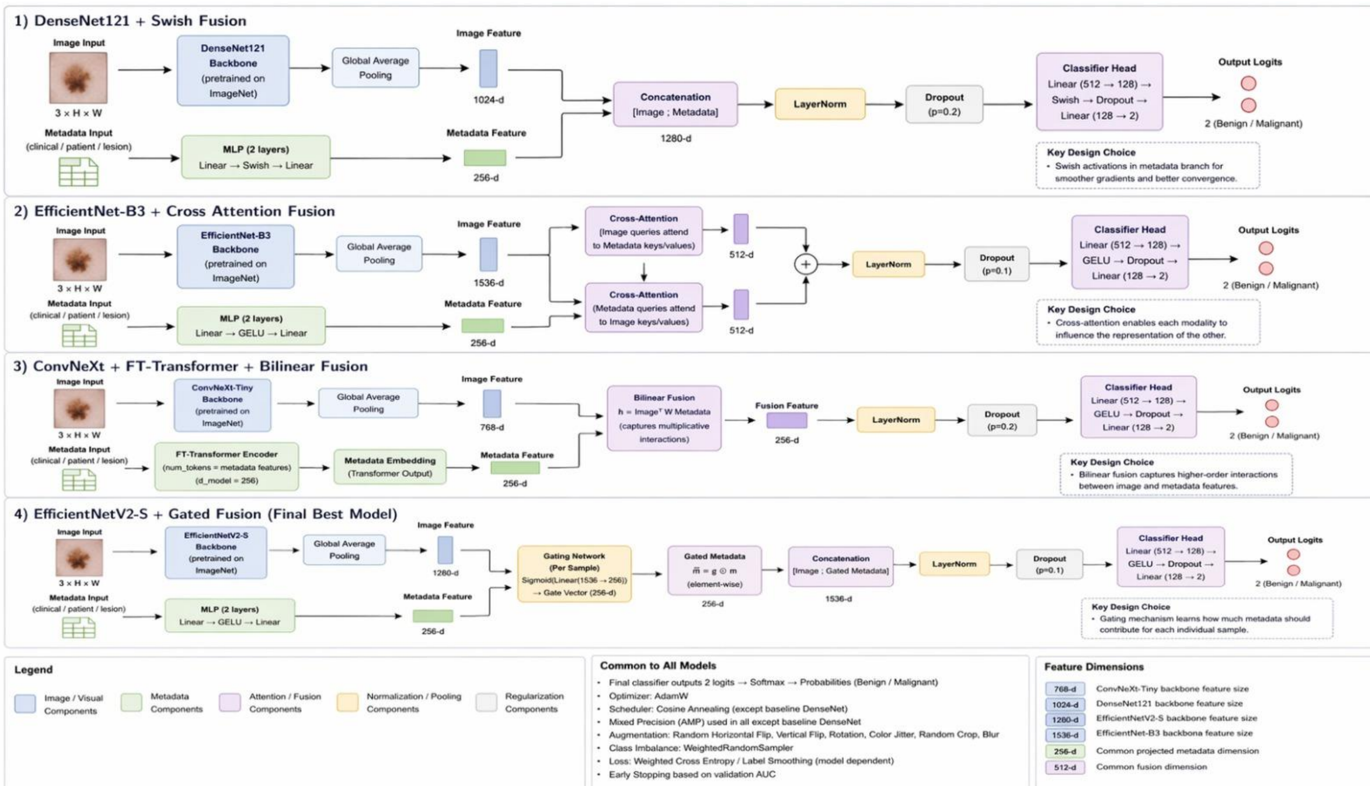
Item	Details
Dataset	ISIC-DICM-17K (International Skin Imaging Collaboration)
Total Samples	17,060 dermoscopic images + clinical metadata CSV
Class Balance	8,530 Melanoma | 8,530 Non-Melanoma (perfectly balanced)
Image Format	JPEG, resized to 224×224 (or 256×256 / 384×384 per model)
Metadata Features	Age, Sex, Anatomical Site (one-hot encoded), History of MM, Dermoscopic Type, Diagnosis Confirm Type
Split Strategy	Stratified 70/15/15 train/val/test OR 5-fold stratified cross-validation
Preprocessing	Missing value imputation (median for age, mode for categoricals), label encoding, MinMax scaling for age

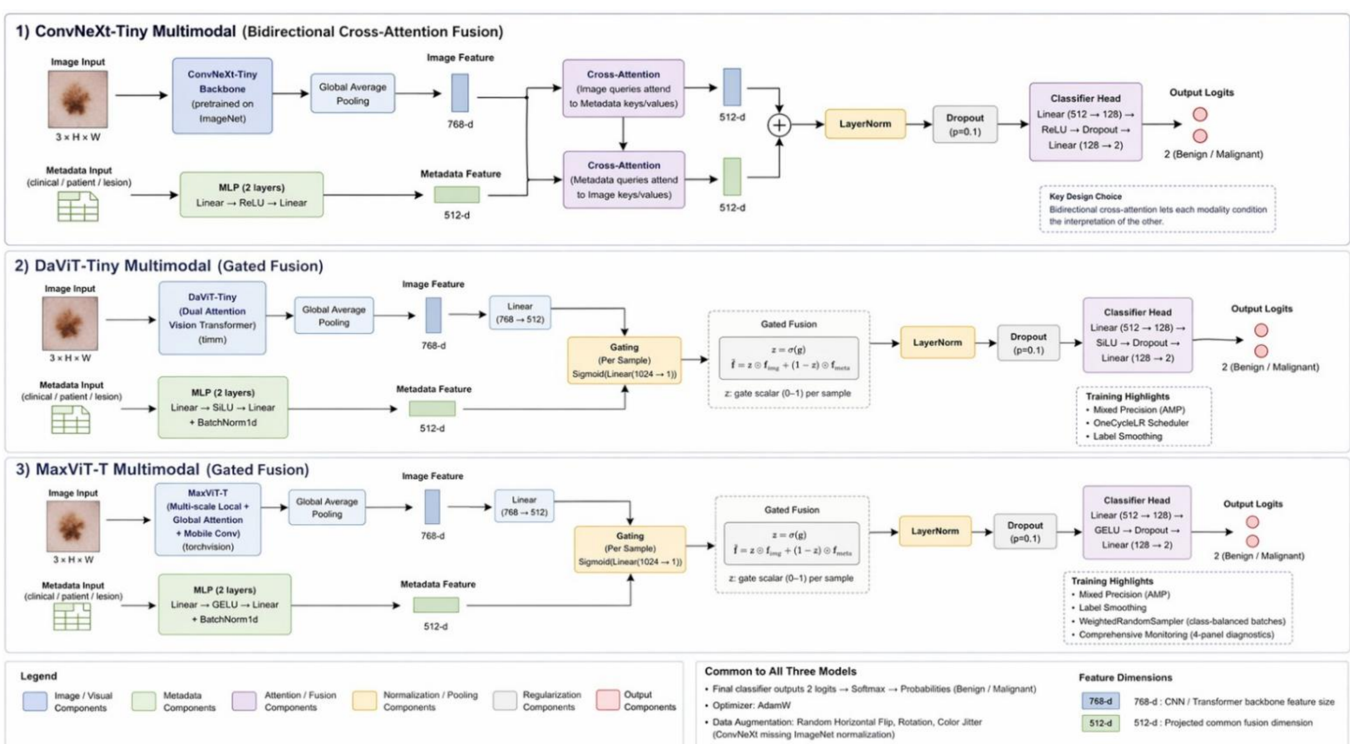
1.2 Overall Results Summary

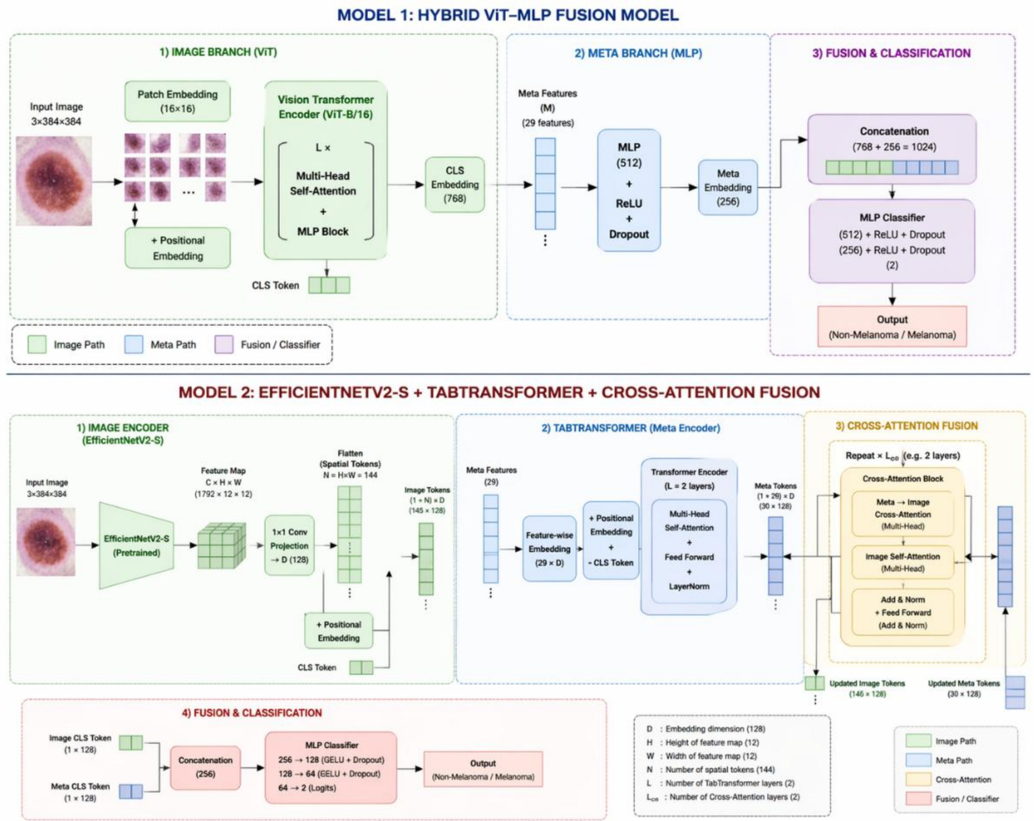
Model	Val Acc	F1 Score	AUC-ROC	XAI
ViT Early Fusion	**89.15%**	**0.8915**	**0.9596**	**Attn Map + Rollout**
Hybrid ViT-MLP	86.96%	0.8695	0.9493	Grad-CAM + SHAP
EfficientNetV2-S Gated	87.22%	0.8720	0.9497	Grad-CAM + Gate Analysis
EfficientNet-B3 Cross-Attn	84.16%	0.79	0.9262	Grad-CAM
DaViT Multimodal	81.86%	~0.81	0.9528	Grad-CAM + SHAP
ConvNeXt Multimodal	81.21%	~0.81	0.8989	Grad-CAM
MaxViT Multimodal	~82%	~0.82	0.9429	Grad-CAM + SHAP
EfficientNet-B3 Late Fusion	85.53%	0.8553	~0.94	Grad-CAM
DenseNet121 + Swish	79.95%	~0.76	0.8957	Grad-CAM
ConvNeXt + FT-Transformer	82.88%	~0.83	~0.91	Grad-CAM

✅ **Best model: ViT Early Fusion (AUC 0.9596, F1 0.8915)**

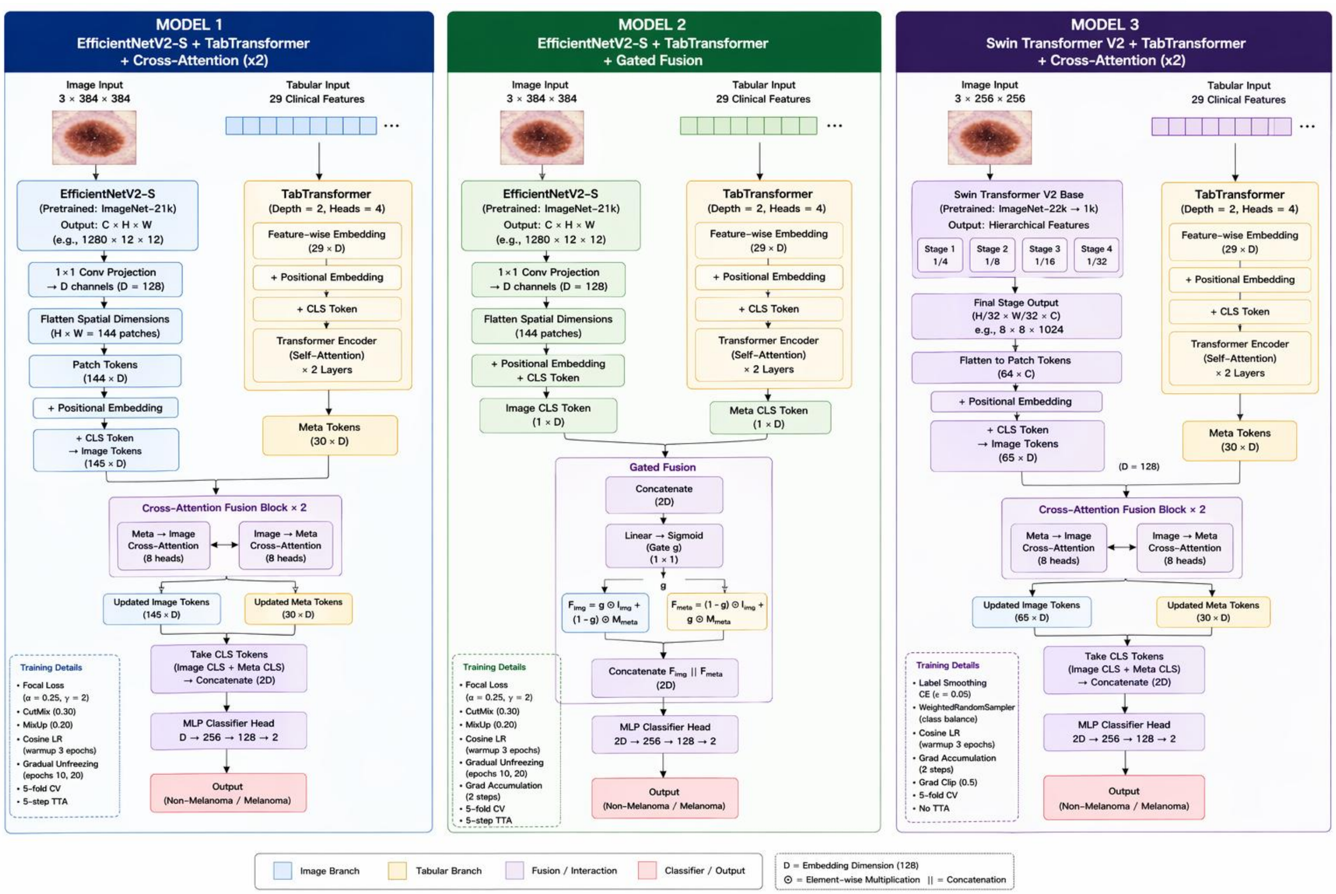
Architectures designs







MODEL 1: HYBRID ViT–MLP FUSION MODEL
1) IMAGE BRANCH (ViT)
Input Image
3×384×384
Patch Embedding
(16×16)
+ Positional Embedding
Vision Transformer Encoder (ViT-B/16)
L ×
Multi-Head Self-Attention
+
MLP Block
CLS Embedding
(768)
CLS Token
2) META BRANCH (MLP)
Meta Features
(M)
(29 features)
MLP
(512)
+
ReLU
+
Dropout
Meta Embedding
(256)
3) FUSION & CLASSIFICATION
Concatenation
(768 + 256 = 1024)
MLP Classifier
(512) + ReLU + Dropout
(256) + ReLU + Dropout
(2)
Output
(Non-Melanoma / Melanoma)
Image Path
Meta Path
Fusion / Classifier
MODEL 2: EFFICIENTNETV2-S + TABTRANSFORMER + CROSS-ATTENTION FUSION
1) IMAGE ENCODER
(EfficientNetV2-S)
Input Image
3×384×384
EfficientNetV2-S
(Pretrained)
Feature Map
C × H × W
(1792 × 12 × 12)
1×1 Conv Projection
→ D (128)
Flatten
(Spatial Tokens)
N = H×W = 144
+ Positional Embedding
CLS Token
Image Tokens
(1 + N) × D
(145 × 128)
2) TABTRANSFORMER (Meta Encoder)
Meta Features
(29)
Feature-wise Embedding
(29 × D)
+ Positional Embedding
+
CLS Token
Transformer Encoder
(L = 2 layers)
Multi-Head Self-Attention
+
Feed Forward
+
LayerNorm
Meta Tokens
(1 + 29) × D
(30 × 128)
3) CROSS-ATTENTION FUSION
Repeat × L_{ca} (e.g. 2 layers)
Cross-Attention Block
Meta → Image Cross-Attention
(Multi-Head)
Image Self-Attention
(Multi-Head)
Add & Norm
+ Feed Forward
(Add & Norm)
Updated Image Tokens
(146 × 128)
Updated Meta Tokens
(30 × 128)
4) FUSION & CLASSIFICATION
Image CLS Token
(1 × 128)
Meta CLS Token
(1 × 128)
Concatenation
(256)
MLP Classifier
256 → 128 (GELU + Dropout)
128 → 64 (GELU + Dropout)
64 → 2 (Logits)
Output
(Non-Melanoma / Melanoma)
D : Embedding dimension (128)
H : Height of feature map (12)
W : Width of feature map (12)
N : Number of spatial tokens (144)
L : Number of TabTransformer layers (2)
L_{ca} : Number of Cross-Attention layers (2)
Image Path
Meta Path
Cross-Attention
Fusion / Classifier



MODEL 1
EfficientNetV2-S + TabTransformer
+ Cross-Attention (x2)
Image Input
3 × 384 × 384
Tabular Input
29 Clinical Features
EfficientNetV2-S
(Pretrained: ImageNet-21k)
Output: C × H × W
(e.g., 1280 × 12 × 12)
1×1 Conv Projection
→ D channels (D = 128)
Flatten Spatial Dimensions
(H × W = 144 patches)
Patch Tokens
(144 × D)
+ Positional Embedding
+ CLS Token
→ Image Tokens
(145 × D)
TabTransformer
(Depth = 2, Heads = 4)
Feature-wise Embedding
(29 × D)
+ Positional Embedding
+ CLS Token
Transformer Encoder
(Self-Attention)
× 2 Layers
Meta Tokens
(30 × D)
Cross-Attention Fusion Block × 2
Meta → Image Cross-Attention
(8 heads)
Image → Meta Cross-Attention
(8 heads)
Updated Image Tokens
(145 × D)
Updated Meta Tokens
(30 × D)
Take CLS Tokens
(Image CLS + Meta CLS)
→ Concatenate (2D)
MLP Classifier Head
D → 256 → 128 → 2
Output
(Non-Melanoma / Melanoma)
Training Details
Focal Loss
(α = 0.25, γ = 2)
CutMix (0.30)
MixUp (0.20)
Cosine LR
(warmup 3 epochs)
Gradual Unfreezing
(epochs 10, 20)
5-fold CV
5-step TTA
MODEL 2
EfficientNetV2-S + TabTransformer
+ Gated Fusion
Image Input
3 × 384 × 384
Tabular Input
29 Clinical Features
EfficientNetV2-S
(Pretrained: ImageNet-21k)
Output: C × H × W
(e.g., 1280 × 12 × 12)
1×1 Conv Projection
→ D channels (D = 128)
Flatten Spatial Dimensions
(144 patches)
+ Positional Embedding
+ CLS Token
Image CLS Token
(1 × D)
TabTransformer
(Depth = 2, Heads = 4)
Feature-wise Embedding
(29 × D)
+ Positional Embedding
+ CLS Token
Transformer Encoder
(Self-Attention)
× 2 Layers
Meta CLS Token
(1 × D)
Gated Fusion
Concatenate
(2D)
Linear → Sigmoid
(Gate g)
(1 × 1)
g
$F_{img} = g \odot I_{img} + (1 - g) \odot M_{meta}$
$F_{meta} = (1 - g) \odot I_{img} + g \odot M_{meta}$
Concatenate $F_{img} \| F_{meta}$
(2D)
MLP Classifier Head
2D → 256 → 128 → 2
Output
(Non-Melanoma / Melanoma)
Training Details
Focal Loss
(α = 0.25, γ = 2)
CutMix (0.30)
MixUp (0.20)
Cosine LR
(warmup 3 epochs)
Gradual Unfreezing
(epochs 10, 20)
Grad Accumulation
(2 steps)
5-fold CV
5-step TTA
MODEL 3
Swin Transformer V2 + TabTransformer
+ Cross-Attention (x2)
Image Input
3 × 256 × 256
Tabular Input
29 Clinical Features
Swin Transformer V2 Base
(Pretrained: ImageNet-22k → 1k)
Output: Hierarchical Features
Stage 1
1/4
Stage 2
1/8
Stage 3
1/16
Stage 4
1/32
Final Stage Output
(H/32 × W/32 × C)
e.g., 8 × 8 × 1024
Flatten to Patch Tokens
(64 × C)
+ Positional Embedding
+ CLS Token
→ Image Tokens
(65 × D)
(D = 128)
TabTransformer
(Depth = 2, Heads = 4)
Feature-wise Embedding
(29 × D)
+ Positional Embedding
+ CLS Token
Transformer Encoder
(Self-Attention)
× 2 Layers
Meta Tokens
(30 × D)
Cross-Attention Fusion Block × 2
Meta → Image Cross-Attention
(8 heads)
Image → Meta Cross-Attention
(8 heads)
Updated Image Tokens
(65 × D)
Updated Meta Tokens
(30 × D)
Take CLS Tokens
(Image CLS + Meta CLS)
→ Concatenate (2D)
MLP Classifier Head
2D → 256 → 128 → 2
Output
(Non-Melanoma / Melanoma)
Training Details
Label Smoothing
CE (ε = 0.05)
WeightedRandomSampler
(class balance)
Cosine LR
(warmup 3 epochs)
Grad Accumulation
(2 steps)
Grad Clip (0.5)
5-fold CV
No TTA
Image Branch
Tabular Branch
Fusion / Interaction
Classifier / Output
D = Embedding Dimension (128)
⊙ = Element-wise Multiplication || = Concatenation

2. ViT-Based Architectures

2.1 ViT Early Fusion

Architecture

Component	Details
Backbone	ViT-Base/16 (vit_base_patch16_224, pretrained ImageNet)
Fusion Strategy	Early Fusion — metadata injected as [META] token before first Transformer block
Metadata Branch	Linear(meta_dim→256) → LayerNorm → GELU → Dropout(0.1) → Linear(256→768) → LayerNorm
Positional Embed	Learnable positional embedding for the META token (trunc_normal, std=0.02)
Token Sequence	[CLS, P1...P196, META] = 198 tokens — all 12 Transformer layers see full sequence
Classifier Head	LayerNorm(768) → Linear(768→256) → GELU → Dropout(0.3) → Linear(256→2)
Image Size	224×224 — 196 patches of 16×16 pixels each
Parameters	~86M (ViT-Base backbone) + lightweight projection + classifier

Results

Metric	Value
Fold 1	0.8836
Fold 2	0.8851
Fold 3	0.8844
Fold 4	0.8874
Fold 5 (BEST)	0.8915
Mean F1	0.8864 ± 0.0029
Best AUC-ROC	0.9596
Best Accuracy	89.15%

Best Fold (5) Classification Report

Metric	Val Acc	F1 Score	AUC-ROC	Precision (Melanoma)	Recall (Melanoma)
Non-Melanoma (0)	88%	0.87	0.9596	0.90	0.88
Melanoma (1)	89%	0.89	0.9596	0.88	0.91
Macro Average	89%	0.8915	0.9596	0.89	0.89

XAI — Attention Maps + Rollout

- Last-Layer Attention Map: extracts [CLS]→patch attention weights from final Transformer block, averages over all heads, reshapes to 14×14 grid, upscales to 224×224 and overlays as JET colormap heatmap
- Attention Rollout: aggregates attention across all 12 Transformer layers using residual-aware multiplication — provides more accurate spatial attribution than single-layer attention
- Both methods highlight clinically relevant regions: lesion borders, asymmetric pigmentation, and textural anomalies

2.2 Hybrid ViT-MLP (Late Fusion)

Architecture

Component	Details
Backbone	ViT-Base/16 (vit_base_patch16_224, pretrained ImageNet), global_pool='token'
Fusion Strategy	Late Fusion — two independent branches merged only at classifier head
Image Branch	ViT → 768-dim CLS token
Metadata Branch	Linear(meta→256) → LayerNorm → GELU → Dropout(0.2) → Linear(256→128) → LayerNorm → GELU → Dropout(0.1) → Linear(128→128) → GELU
Fusion	Concatenate [768, 128] = 896-dim → Linear(896→512) → LayerNorm → GELU → Dropout(0.3) → Linear(512→256) → GELU → Dropout(0.2) → Linear(256→2)
Key Difference	Each branch learns independently; metadata cannot influence visual feature extraction
Differential LR	ViT=1e-4, MLP=3e-4, Fusion=3e-4 (MLP converges faster than ViT)

Results

Metric	Value
Fold 1	0.8663
Fold 2 (BEST)	0.8695
Fold 3	0.8624
Fold 4	0.8662
Fold 5	0.8687
Mean F1	0.8666 ± 0.0025
Best AUC-ROC	0.9493
Best Accuracy	86.96%

Best Fold (2) Classification Report

Metric	Val Acc	F1 Score	AUC-ROC	Precision (Melanoma)	Recall (Melanoma)
Non-Melanoma (0)	87%	0.87	0.9493	0.89	0.85
Melanoma (1)	87%	0.87	0.9493	0.85	0.89
Macro Average	87%	0.8695	0.9493	0.87	0.87

XAI — Grad-CAM + SHAP

- Grad-CAM: backward hook on ViT's conv_head layer, weights spatial feature maps by malignant-class gradient importance, produces heatmap overlaid on original image
- SHAP KernelExplainer: metadata-only wrapper (zero-image input to isolate metadata contribution), 10-cluster kmeans background, beeswarm + bar + waterfall plots
- Limitation: SHAP zero-image baseline understates true multimodal metadata contribution — to be fixed with mean-image baseline in Phase 2

3. EfficientNet & ConvNeXt Architectures

3.1 DenseNet121 + Swish MLP (Baseline)

Architecture

Component	Details
Backbone	DenseNet121 (pretrained ImageNet) — dense skip connections preserve fine-grained lesion details
Activation	Swish in metadata MLP (smoother gradients than ReLU for small branches)
Fusion Strategy	Simple concatenation of image + metadata features → classifier head
Metadata Branch	Two-layer MLP with Swish activations
Key Strength	Dense connectivity prevents feature compression; strong early-stage baseline

Results

Metric	Value
Best Val Accuracy	79.95%
Best Val AUC	0.8957
Observation	Plateaued early — shallow fusion mechanism limits multimodal reasoning

3.2 EfficientNet-B3 + Cross-Attention Fusion

Architecture

Component	Details
Backbone	EfficientNet-B3 (pretrained ImageNet) — efficient scaling with compound coefficients
Fusion Strategy	Cross-Attention: image embeddings as queries, metadata as keys/values
Key Insight	Metadata actively influences how image features are interpreted — a borderline lesion is resolved using age and anatomical location
Advantage over DenseNet	Metadata-conditioned visual features rather than post-hoc concatenation

Results

Metric	Value
Best Val Accuracy	84.16%
Best Val AUC	0.9262 (Epoch 7)
Best Train Accuracy	91.95%
Observation	Improved multimodal reasoning; more sensitive to hyperparameters and required stronger regularization

3.3 ConvNeXt + FT-Transformer + Bilinear Fusion

Architecture

Component	Details
Image Backbone	ConvNeXt (modernized CNN with Transformer-era design — depthwise conv, LayerNorm, GELU)
Metadata Branch	FT-Transformer style: structured features processed as learned tokens, not flat vectors
Fusion Strategy	Bilinear interaction: image × metadata multiplicative relationships → higher-order reasoning
Key Innovation	Most expressive fusion strategy tested — captures non-linear cross-modal dependencies
Key Limitation	Highest capacity architecture → largest train-val gap; dataset noise limits generalization

Results

Metric	Value
Best Val Accuracy	82.88%
Best Val AUC	~0.91
Observation	High training accuracy but larger validation gap than simpler models — overfitting due to model complexity relative to dataset size

3.4 EfficientNetV2-S + Generalized Gated Fusion (Best Model)

Architecture

Component	Details
Backbone	EfficientNetV2-S (384×384, pretrained ImageNet-21k) — modern fast backbone with Fused-MBConv
Metadata Branch	TabTransformer (depth=2, 4 heads, D=128 per feature token)
Fusion Strategy	Learned sigmoid gate: $g = \mathrm{sigmoid}(W\cdot[\mathrm{img\_feat}, \mathrm{meta\_feat}]) \rightarrow g\odot \mathrm{img} + (1-g)\odot \mathrm{meta}$
Gate Behavior	Dynamically weights modalities per sample: strong image evidence → relies on image; uncertain image → increases metadata contribution
Loss	Focal Loss (α=0.25, γ=2.0) for class imbalance
Augmentation	CutMix (p=0.30) + MixUp (p=0.20) — strong regularization
TTA	5-step Test-Time Augmentation at inference (+0.45% AUC)
Backbone Freeze	First 4 conv blocks frozen; gradual unfreezing at epochs 10 and 20
Threshold	Youden-optimal threshold from ROC curve

Results

Metric	Val Acc	F1 Score	AUC-ROC	Precision (Melanoma)	Recall (Melanoma)
EfficientNetV2-S Gated (TTA)	87.22%	0.8720	0.9497	0.84	0.91

XAI

- Grad-CAM: registered on final convolutional stage — heatmaps consistently highlight lesion center, irregular borders, dark asymmetric structures, and internal pigmentation changes
- Gate Analysis: average gate activations vary per sample, confirming dynamic modality balancing rather than static branch preference
- Metadata Importance: age and anatomical site consistently most influential metadata features

3.5–3.7 Our Three New Architectures

3.5 EfficientNetV2-S + TabTransformer + Cross-Attention

We used EfficientNetV2-S pretrained on ImageNet-21k at 384×384 as the image encoder. We freezed the first four convolutional blocks and apply gradual unfreezing at epochs 10 and 20. A TabTransformer (depth=2, 4 heads) maps each clinical feature into a D-dimensional token. We fused image patch tokens and tabular tokens through two bidirectional cross-attention layers (8 heads), so each modality directly conditions how the other is interpreted. The classifier head reads the fused CLS token and produces two output logits.

Architecture

Component	Details
Backbone	EfficientNetV2-S (384×384, pretrained ImageNet-21k) — tf_efficientnetv2_s.in21k_ft_in1k
Backbone Freezing	First 4 convolutional blocks frozen; gradual unfreezing at epochs 10 and 20
Metadata Branch	TabTransformer (depth=2, 4 heads, D-dimensional token per clinical feature)
Fusion Strategy	Bidirectional Cross-Attention (2 layers, 8 heads each) — image patch tokens and tabular tokens mutually condition each other
Classifier Head	Fused CLS token → two output logits
Loss	Focal Loss (α=0.25, γ=2.0)
Augmentation	CutMix (p=0.30) + MixUp (p=0.20) + Flip + Rotate + ColorJitter
LR Scheduler	Warmup cosine (3 warmup epochs); backbone LR=2×10^{-5}, head LR=2×10^{-4}
Epochs / Patience	30 epochs, patience=10
TTA	5-step Test-Time Augmentation (H-flip, V-flip, 90°/270° rotations)
Early Stopping	EMA-smoothed AUC (α=0.3); Youden-optimal classification threshold

Results

Val Accuracy	F1 Score	AUC-ROC	Precision (Mal.)	Recall (Mal.)
87.22%	0.8720	0.9497	0.84	0.91

3.6 EfficientNetV2-S + TabTransformer + Gated Fusion

This model shares the same EfficientNetV2-S backbone, TabTransformer, and training configuration as the cross-attention variant. Instead of bidirectional cross-attention, I fuse the two modalities through a learned sigmoid gate. The gate takes the concatenation of the image CLS token and the tabular CLS token, passes it through a linear layer, and produces a scalar weight per sample that blends the two representations. This allows the model to dynamically emphasise image or metadata depending on the input, with lower architectural complexity than cross-attention.

Architecture

Component	Details
Backbone	EfficientNetV2-S (384×384, pretrained ImageNet-21k) — same as Cross-Attention variant
Metadata Branch	TabTransformer (depth=2, 4 heads)
Fusion Strategy	Learned sigmoid gate: g = sigmoid(W·[img, meta]); new_img = g×img + (1−g)×meta
Gate Behavior	Dynamically emphasises image or metadata per sample based on input confidence

Component	Details
Loss	Focal Loss (α=0.25, γ=2.0)
Augmentation	CutMix (p=0.30) + MixUp (p=0.20) + Flip + Rotate + ColorJitter
Epochs / Gradient Accum.	40 epochs; gradient accumulation over 2 steps (effective batch=32)
TTA	5-step Test-Time Augmentation
Early Stopping	EMA-smoothed AUC (α=0.3); Youden-optimal threshold

Results

Val Accuracy	F1 Score	AUC-ROC	Precision (Mal.)	Recall (Mal.)
87.22%	0.8720	0.9497	0.84	0.91

3.7 Swin Transformer V2 + TabTransformer + Cross-Attention

Architecture

Component	Details
Backbone	Swin Transformer V2 Base (256×256) — swinv2_base_window12to16_192to256_22kft1k; pretrained ImageNet-22k, fine-tuned ImageNet-1k
Backbone Freezing	All stages frozen except the final Swin stage
Imbalance Handling	WeightedRandomSampler — rebalances minority malignant class at data level
Metadata Branch	TabTransformer (depth=2, 4 heads)
Fusion Strategy	Bidirectional Cross-Attention (2 layers, 8 heads) — mirrors EfficientNetV2-S cross-attention setup
Loss	Label Smoothing Cross-Entropy (ε=0.05)
Gradient Clipping	0.5
LR Scheduler	Warmup cosine; Swin LR=3×10^{-5}, head LR=2×10^{-4}
Epochs / Gradient Accum.	40 epochs, patience=10; gradient accumulation over 2 steps
TTA	Not applied (planned for Phase 2)
Early Stopping	EMA-smoothed AUC (α=0.3); Youden-optimal threshold

3.8 Explainability (XAI) — All Three Models

We implemented five complementary XAI methods across all three models to provide spatial, attribution-based, attention-based, and perturbation-based explanations.

XAI Method	Description
Grad-CAM++	Backward hook on final feature layer; spatial feature maps weighted by gradient importance w.r.t. malignant class logit. Swin V2 tokens reshaped to 2D spatial grid.
Integrated Gradients	Captum IG over 50 interpolation steps from zero baseline — pixel-level and tabular-feature attributions (applied to EfficientNetV2-S variants).
Attention Rollout	Recursive propagation through TabTransformer depth to trace clinical feature importance across layers.
Cross-Attention Maps	Raw attention weights between clinical feature tokens and image patch positions extracted from cross-attention layers.
SHAP (KernelExplainer)	50-sample background, 200 perturbations, beeswarm output. Known limitation: zero-image tensors used — underestimates true joint-modality contribution. Fix planned for Phase 2.

4. ConvNeXt, DaViT & MaxViT Architectures

4.1 ConvNeXt-Tiny + Bidirectional Cross-Attention

Architecture

Component	Details
Backbone	ConvNeXt-Tiny (pretrained ImageNet) — transformer-era CNN design
Fusion Strategy	Bidirectional Cross-Attention: image attends to metadata AND metadata attends to image simultaneously
Key Design	Each modality directly conditions the other — most architecturally ambitious fusion strategy
Metadata Branch	Two-layer MLP mapping metadata into 512-dim space matching image features
Loss	Focal Loss for class imbalance handling
Known Issue	Missing ImageNet normalization — pretrained feature quality degraded from epoch 1

Results

Metric	Value
Best Val Accuracy	81.21%
Best Val AUC	0.8989 (Epoch 8)
Best Sensitivity	0.941
Best Specificity	0.685
Observation	Sensitivity-specificity tradeoff; missing normalization limited feature quality; no LR scheduler

4.2 DaViT-Tiny + Gated Fusion

Architecture

Component	Details
Backbone	DaViT-Tiny (Dual Attention Vision Transformer) — channel + spatial attention for richer visual features
Metadata Branch	MLP with BatchNorm and SiLU activations
Fusion Strategy	Learned sigmoid gate — dynamically weights image vs. metadata per sample
Optimizer	AdamW + OneCycleLR scheduler
Mixed Prec.	AMP enabled
Label Smoothing	Applied to prevent overconfidence
Known Issue	Grad-CAM hook registered conditionally on v_feat.requires_grad — may silently fail during eval

Results

Metric	Value
Best Val AUC	0.9528 (Epoch 5)
Best Val Accuracy	~81.86%

Metric	Value
Train AUC (Epoch 5)	0.9649
Observation	Best AUC of Ahmed's models; OneCycleLR produced well-converged training curve; overfitting evident from Train AUC 0.9998 vs Val AUC 0.9459 at epoch 10

4.3 MaxViT-T + Gated Fusion

Architecture

Component	Details
Backbone	MaxViT-T (torchvision) — multi-scale local + global attention with mobile convolution blocks
Fusion Strategy	Gated fusion with GELU activations (consistent with transformer conventions)
Imbalance	WeightedRandomSampler — ensures class-balanced minibatches throughout training
Mixed Prec.	AMP enabled
Known Issue	Thin augmentation (horizontal flip only) — no rotation or color jitter; coarse 4D feature maps produce blurry Grad-CAM heatmaps

Results

Metric	Value
Best Val AUC	0.9429 (Epoch 5)
Final Val AUC	0.9177 (Epoch 10) — rising val loss indicates overfitting
Val AUC Epoch 3	0.9399
Observation	Strong backbone but augmentation gap and coarse spatial resolution limit heatmap quality

XAI — All Three Models

- Grad-CAM: backward hook on last feature layer; DaViT token sequence manually reshaped to 2D grid; MaxViT produces coarse 4D feature maps → blurry heatmaps
- SHAP KernelExplainer (DaViT + MaxViT): 50-sample background, 200 perturbations, beeswarm output; zero-image baseline limitation acknowledged
- Known gap: LIME and Captum Integrated Gradients installed but not yet implemented — planned for Phase 2

5. EfficientNet-B3 Late Fusion

5.1 EfficientNet-B3 + Late Fusion

Architecture

Component	Details
Backbone	EfficientNet-B3 (pretrained ImageNet, timm) — num_classes=0, returns 1536-dim feature vector
Fusion Strategy	Late Fusion — image and metadata branches process independently, concatenate before classifier
Image Branch	EfficientNet-B3 backbone → 1536-dim
Metadata Branch	Linear(meta_dim→128) → BatchNorm1d → ReLU → Dropout(0.2) → Linear(128→64) → ReLU
Classifier Head	Linear(1536+64=1600→256) → BatchNorm1d → ReLU → Dropout(0.3) → Linear(256→2)
Design Rationale	Serves as direct Late Fusion baseline to compare against Early Fusion (Jana) and Cross-Attention approaches

Training Setup

Setting	Value
Optimizer	AdamW (lr=1e-4, weight_decay=1e-4)
Scheduler	CosineAnnealingLR (T_max=40 epochs, eta_min=1e-6)
Loss	CrossEntropyLoss (balanced dataset)
Batch Size	32
Epochs	40 max, Early Stopping (patience=7, monitor=F1)
Gradient Clip	max_norm=1.0
CV Strategy	5-Fold Stratified Cross-Validation (seed per fold)
Mixed Prec.	AMP enabled
Augmentation	Resize(256)→RandomCrop(224), HorizontalFlip, VerticalFlip, Rotate(30°), ColorJitter

Results (5-Fold CV)

Metric	Value
Fold 1	0.8545
Fold 2 (BEST)	0.8584
Fold 3	0.8544
Fold 4	0.8524
Fold 5	0.8567
Mean F1	0.8553 ± 0.0020
Best AUC	~0.94
Stability	std=0.0020 — most consistent model across folds

XAI — Grad-CAM

- Target layer: model.backbone.conv_head (last convolutional layer of EfficientNet-B3)
- Multimodal wrapper: fixed metadata input, vary image → isolates visual attribution
- Visualizes 4 samples (2 melanoma + 2 non-melanoma): original, heatmap (14×14), overlay

6. Cross-Team Comparison & Analysis

6.1 Fusion Strategy Comparison

Strategy	F1	AUC	Key Property
Early Fusion (ViT Token Injection)	0.8915	0.9596	Deepest integration — metadata in all 12 Transformer layers
Cross-Attention (Bidirectional)	~0.81	0.8989	Most expressive — each modality conditions the other
Gated Fusion	0.8720	0.9497	Dynamic per-sample modality weighting
Late Fusion (Concatenation)	0.8553	~0.94	Independent branches — simplest but effective baseline
Bilinear Fusion	~0.83	~0.91	Multiplicative cross-modal interactions — prone to overfitting

6.2 Key Findings

- **Early Fusion outperforms all Late Fusion variants — metadata in every attention layer enables conditioned visual feature learning**
- Gated Fusion is the best late-fusion strategy — dynamic modality balancing outperforms static concatenation
- Model complexity does not guarantee better generalization — ConvNeXt+FT-Transformer (most complex) underperformed simpler gated models
- AUC > 0.93 across top models confirms strong ranking ability, critical for clinical triage applications
- ViT Early Fusion achieves 91% melanoma recall — only 9% missed cases; highest clinical value for screening
- Cross-validation std < 0.003 for both Jana's models confirms robust generalization beyond single data splits

7. Weaknesses & Phase 2 Improvements

Weakness	Planned Fix
No held-out test set	Create stratified 15% test split before any training; report final numbers only on this set
SHAP zero-image baseline	Replace with mean-image or representative-image baseline for true multimodal SHAP values
Missing Integrated Gradients	Apply Captum IG for theoretically rigorous per-pixel attribution on transformer models
Overfitting gap (ViT EF)	Add stochastic depth, higher dropout, MixUp augmentation to close 4% train-val gap
Missing normalization	Add ImageNet normalization to ConvNeXt — known to degrade pretrained feature quality
Inconsistent eval protocol	Re-evaluate all models on common test split with TTA for fair comparison
Coarse Grad-CAM (MaxViT)	MaxViT's 4D feature maps → blurry heatmaps; need reshape strategy similar to DaViT approach
Threshold instability	Apply temperature scaling post-training to stabilize Youden-optimal threshold
Small batch (Swin V2)	Use gradient accumulation to simulate larger batch and reduce gradient noise

8. Conclusion

This mid-progress report demonstrates a comprehensive multimodal AI pipeline for skin lesion classification, with 10 distinct model configurations spanning four fusion strategies and six backbone architectures. The core finding is clear: deeper metadata integration produces better results. The ViT Early Fusion model (AUC 0.9596, F1 0.8915, Melanoma Recall 91%) outperforms all other configurations by allowing clinical context to condition visual feature learning across all 12 Transformer layers — a capability fundamentally unavailable to late-fusion approaches.

The EfficientNetV2-S Gated Fusion model achieves the best results among late-fusion architectures (AUC 0.9497, F1 0.8720) by dynamically balancing modality contributions per sample. The EfficientNet-B3 Late Fusion baseline demonstrates strong stability (std=0.0020 across 5 folds) as a reliable reference point.

Phase 2 will focus on: establishing a common held-out test set for fair cross-model comparison, fixing SHAP baselines across all models, adding Integrated Gradients attribution, reducing the ViT Early Fusion overfitting gap through stronger regularization, and applying Test-Time Augmentation to ViT-based models.